Introduction to DBMS (Database Management System)

A DBMS is software that helps us store, manage, and retrieve data easily and safely.

Instead of saving data in files like text files or Excel sheets, we use a database controlled by a DBMS.

A database is a structured collection of data stored in a computer system for easy retrieval and management.

Real-life example:

Student records in college

Bank accounts

Instagram users & their posts.

Need for DBMS

Before DBMS, people used file-based systems. They had many problems.

Problems with File System:

Data duplication:
Same data is stored multiple times in different files, wasting storage space.

No security:
Any user can access or modify data without proper authorization.

Difficult to search data:
Finding required data is slow and complex because there is no proper query system.

No backup & recovery:
If data is lost or the system crashes, there is no way to restore the data.

DBMS solves these problems.

Need of DBMS:

1. Avoids Data Redundancy
   Same data is not stored again.
2. Provides Data Security
   Only authorized users can access data.
3. Easy Data Access
   We can easily fetch data using queries.

4. Backup and Recovery
   Data can be recovered if system crashes.
5. Multi-user Access
   Many users can use data at the same time.

DBMS Architecture

DBMS architecture shows how users, DBMS, and database interact.

Types of DBMS Architecture:

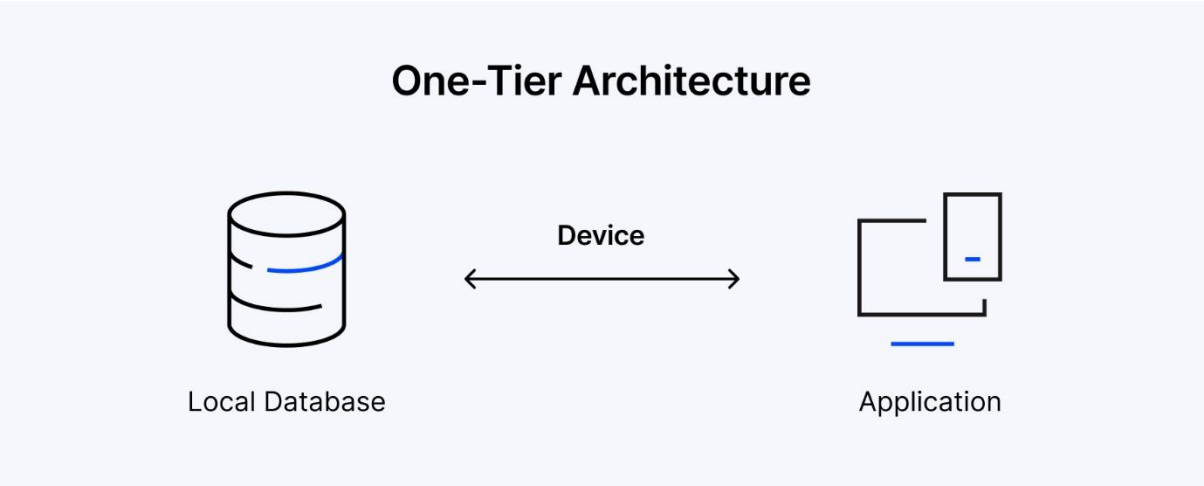
- 1-Tier Architecture
- 2-Tier Architecture
- 3-Tier Architecture

1-Tier Architecture

1-Tier architecture is the simplest DBMS architecture where the user, application, and database are all on the same system. There is no separate server and no network connection involved.

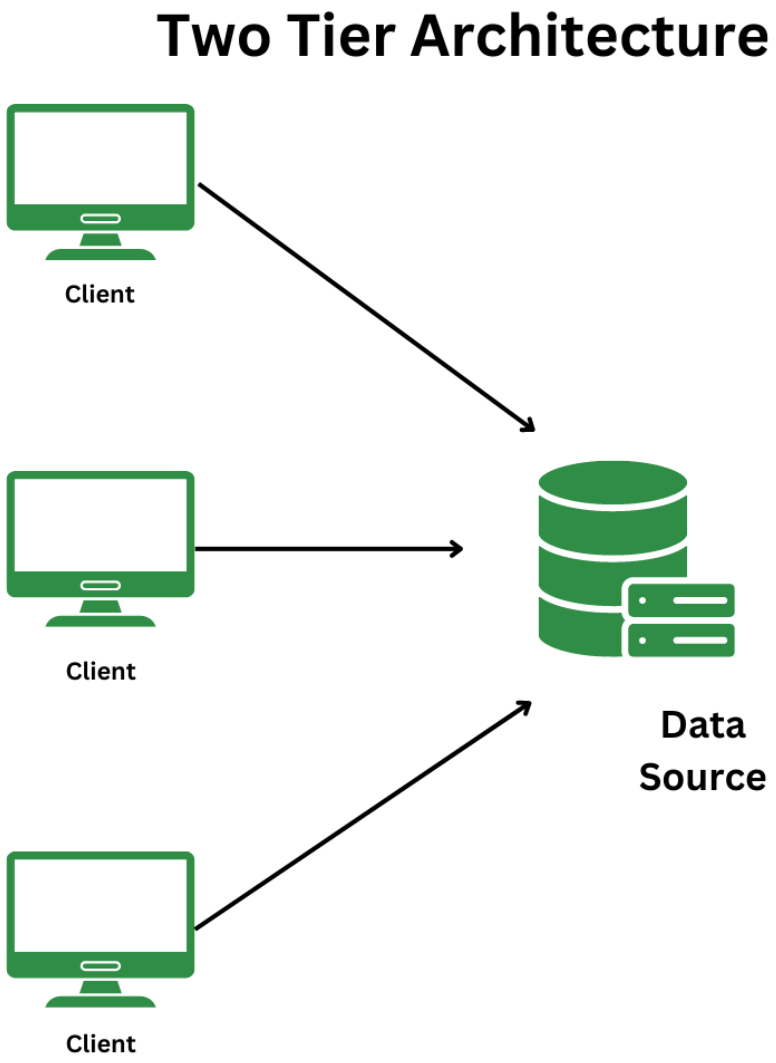
Example:

- Database stored on your computer
- No network involved



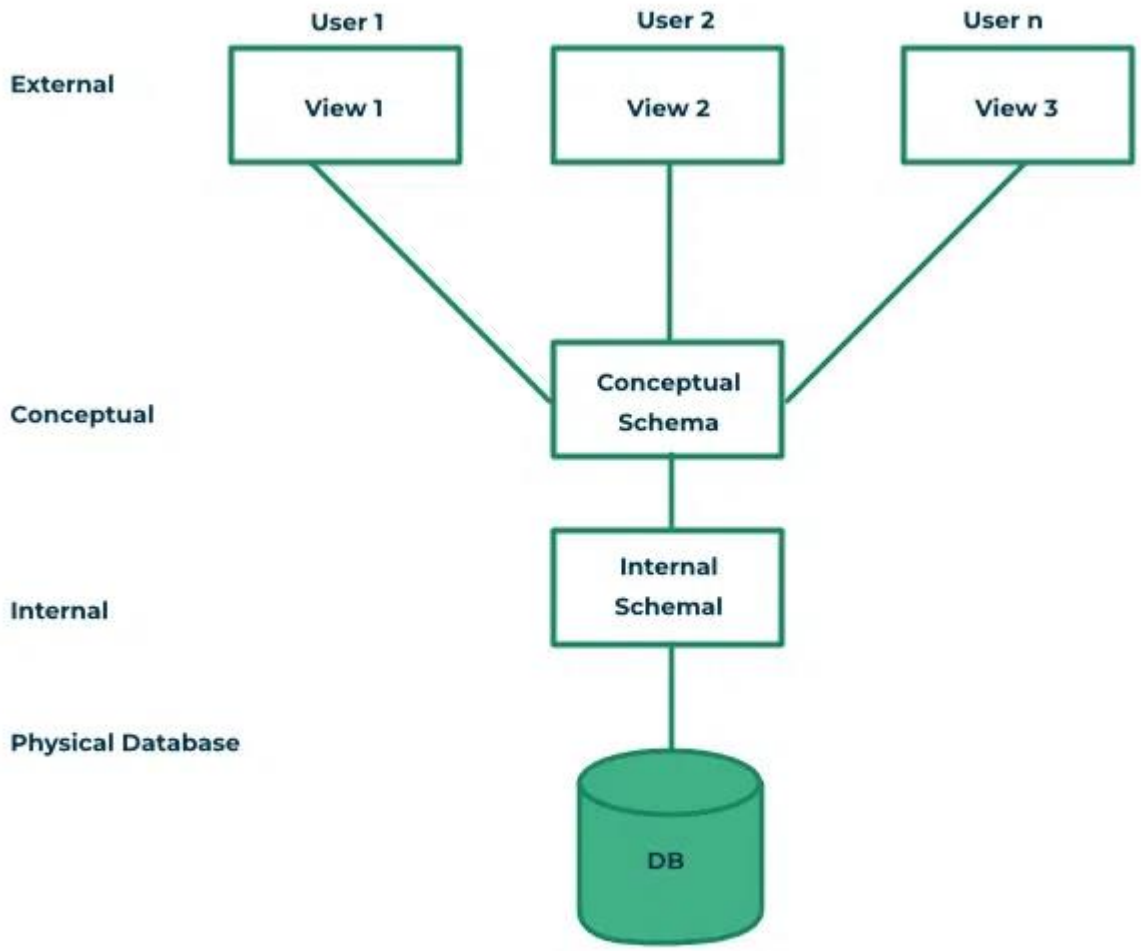
2-Tier Architecture

In 2-tier architecture, the user uses an application (client), and this application directly communicates with the database server to store or retrieve data.



Faster but less secure

3-Tier Architecture



Presentation Layer (Client Layer)

- User interface (website / app)
- Takes input from user

Application Layer (Business Logic Layer)

- Processes user requests
- Applies rules and logic
- Acts as a bridge between user and database.

Database Layer

- Stores and manages actual data.

Example:

- Online banking system
- E-commerce websites
- Social media applications

Why 3-tier is best?

- More secure
- Easy to manage
- Scalable

Database Languages in DBMS

Database languages are used to communicate with the database and perform different operations like creating tables, inserting data, and controlling access.

1. DDL (Data Definition Language)

- Used to define and modify the structure of the database
- It creates, changes, or deletes tables and databases

Operations:

- **CREATE** – Creates database objects like tables
- **ALTER** – Modifies the structure of existing tables
- **DROP** – Deletes tables or databases
- **TRUNCATE** – Removes all records from a table
- **RENAME** – Changes the name of a database object

2. DML (Data Manipulation Language)

- Used to insert, update, and delete data in tables
- Works on the actual data stored in the database

Operations:

- **INSERT** – Adds new data
- **UPDATE** – Modifies existing data
- **DELETE** – Removes data

3. DQL (Data Query Language)

- Used to retrieve or fetch data from the database
- Helps in viewing required data

Operations:

- **SELECT** – Fetches required data

4. DCL (Data Control Language)

- Used to control access and permissions
- Decides who can use or modify the database

Operations:

- **GRANT – Gives permissions to users**
- **REVOKE – Removes permissions from users**

5. TCL (Transaction Control Language)

Used to manage transactions in the database

Ensures data consistency during operations

Operations:

- **COMMIT** – Saves changes permanently
- **ROLLBACK** – Undoes changes
- **SAVEPOINT** – Sets a point to roll back to

DBMS Software

DBMS software is a program that helps users create, store, manage, and retrieve data from a database easily and securely.

It acts as a middle layer between the user and the database, making sure data is organized, safe, and consistent.

MYSQL

MySQL is an open-source relational database management system widely used for web applications. It is fast, lightweight, and easy to learn, which makes it very popular among beginners and startups. MySQL works extremely well with web technologies and is commonly used in dynamic websites and content-driven applications. It is cost-effective, scalable for medium to large systems, and supported by a large developer community.

Oracle

Oracle Database is a commercial, enterprise-level relational database system designed for high performance, security, and reliability. It is commonly used by large organizations for mission-critical applications. Oracle supports advanced features such as high availability, strong transaction control, and complex data handling. It is highly scalable and robust but more complex to manage and significantly more expensive than open-source alternatives.